

Technology Ventures

The teaching team

■ Davide

Lecturer & researcher @ EPFL
(2020-today)

Head of Entrepreneurship
Education Hub (*new!*)

Collaborated w/ startup
accelerators and digital
startups across EU



Carolyn ■

SKACH Program Director
(2021-today)

PhD student at
Entrepreneurship & Tech
Commercialization Chair
(*new!*)

Facilitating tech transfer
between astronomy and
industry

The teaching team

■ Marc

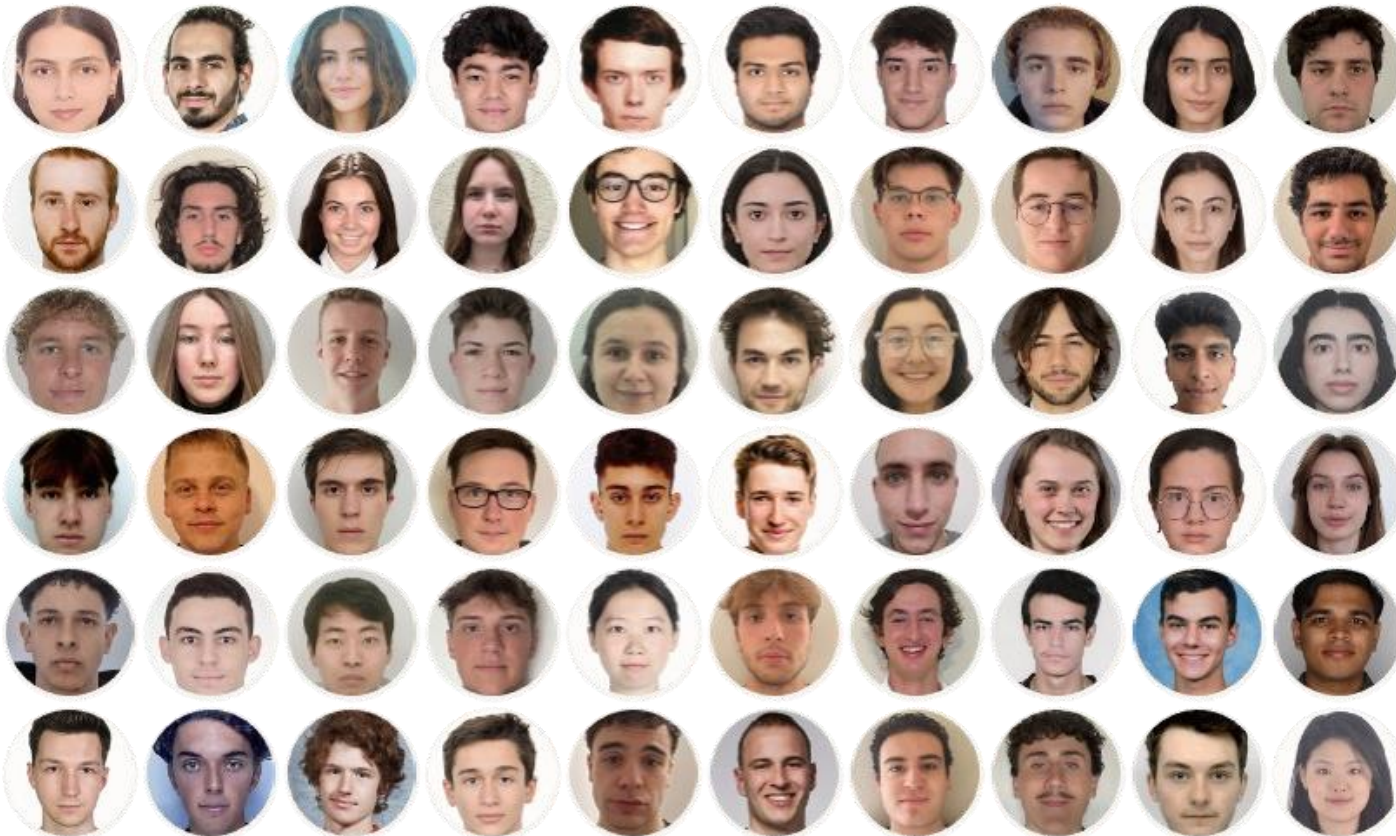
Chair of Entrepreneurship &
Technology Commercialization
@ EPFL (2005-today)

Former VP for Innovation @
EPFL (2017-2021)

Named one of the most
influential researchers in
entrepreneurship (2016)



The student team



60
students

20
masters

27%
women

Entrepreneurship, revisited

*Think of an
entrepreneur*

Name

Characteristic

Entrepreneurship, revisited

- **Occupational identities:** jobs have their “what” and their “who” → including founders of tech ventures
- **Two-way street:** people derive identity from their jobs, and jobs derive their identity from associated people
- **“Glass slipper”:** professions appear “possessed of features that fit certain people yet are improbable for others” (Ashcraft, 2013: 6) leading to occupational segregation (**fits** & **misfits**)
- Today’s class is intended to challenge and reconstruct our (narrow?) view of entrepreneurship



An (unorthodox) introduction

- Introduce & connect
- Course (*personal*) objectives and content
- *Trivia*: the most entrepreneurial nation
- *Trivia*: the valuable startups
- Next steps

ETA
18:15-18:30

PART I

Content & objectives

Introduce & connect



- **Intros (5')**

Who am I

What I am up to these days

Why I am here, my expectations



- **Mind map (10')**

List what the course should cover

Identify 10-15 most important ones

Cluster them



- **Share (15')**

Recap to class
& discuss

Introduce & connect

1

Abdelkefi Ons
Abdelmoula Melik Cyril
Alami-Sounni Kenza Margaux
Anane Rayen
Antonu Jiri

2

Baiges Trilla Roger
Béguin Siméon Jacques
Ben Mansour Takoua
Besnard Mayeul Patrice
Billieux Max

3

Cadorin Léni
Corbière Clémence Priscilla Marie
Costa Roberto Giovanni
Cunningham Sophie Louise
Darbellay Jean-Théo Enrico Valéry

4

Demir Ayça
Djordjevic Luka
Dutruy Eloi
Egri Ecenaz
Fata Jean Marc

5

Fontaine Nicolas
Freiberga Melita
Glaser Philipp
Glauser Loic
Gliga Tatiana

6

González Monfort Pablo
Gonzalez Perez Valeria
Gorgiev David
Grover Tanmay
Hajipour Pariya

7

Huot Adrien Alexandre Marie
Krasna Aljaz
Lefevre Paul Antoine Marie
Maillard Nathan
Martins Piñon Lois

8

Mathieux Gabriel Hugo
Menegalli-Boggelli Patrick
Messin-Roizard Clélia
Mikou Ghita
Modenko Anhelina

9

Moraes De Araujo Pedro Henrique
Moudden Ismail
Okura Ko
Pavan Giovanni
Qi Yifei

10

Richtr Jan
Robert Halpern Luis
Romdhane Mohamed Khalil
Sampaio Alexandre Mattheus
Sharma Tinku

11

Shtilkind Maksim
Silvestre Jules Pierre Ulysse
Sobkowski Lionel
Tekaya Ismael
Tiberi Sorignani Filippo Maria

12

Widmer Yves Denis
Wiletsky Abraham Fischer
Wülser Marc-Antoine
Würigler Raphael
Zhou Zhiqian



- editor of *Student* magazine (1966)

A student founder **story**



- editor of *Student* magazine (1966)

abolition of Resale Price Maintenance Agreements in the UK

- selling records by mail-order (1970)



- editor of *Student* magazine (1966)

abolition of Resale Price Maintenance Agreements in the UK

- selling records by mail-order (1970)
- opening a record shop (1971)
- opening a recording studio (1972)
- launching a record label (1973)





- editor of *Student* magazine (1966)

abolition of Resale Price Maintenance Agreements in the UK

- selling records by mail-order (1970)
- opening a record shop (1971)
- opening a recording studio (1972)
- launching a record label (1973)



The founder- **opportunity** nexus

- **personal development:** identity, experience, talent, mindset, resources
- **opportunity discovery:** identification, evaluation, validation
- **venture building:** business model, resource mobilization, development, entry strategy

Your journey

Date	Topics	Guests	Structure
Sep 9	1. Founder-opportunity nexus		
Sep 16	2. <i>Clinic: ideation and creativity</i>		
Sep 23	3. Opportunities from technology		
Sep 30	4. Opportunities for society		
Oct 7	5. Presentations: opportunity set	Gallo & Conti (Startup Launchpad), Lattuada (Technology Transfer Office), Delisle (SPOT) NEW!	1. Identification (weeks 1-5)
Oct 14	6. Validation I: scientific approach		
Oct 21	--- no session ---		
Oct 28	7. Validation II: industry analysis		2. Validation (weeks 6-9)
Nov 4	8. <i>The uncomfortable truth</i>	Vincent Bieri (Nextthink co-founder )	
Nov 11	9. Feedback session: validation review	Student founders (tbc)	
Nov 18	10. Business models I: creating value		
Nov 25	11. <i>Field trip: EPFL Investor Day</i>		
Dec 2	12. Business models II: capturing value		3. Business design (weeks 10-14)
Dec 9	13. <i>Clinic: pitching and deck design</i>		
Dec 16	14. Presentation: venture concepts	Giorgia Scetta (MassChallenge), Cecilia Molina (TalentKick), Mike Lamotte (InnovationPark), others tbc	

... to be continued (MGT-436)

Your journey

Date	Topics	Guests	Structure
Sep 9	1. Founder-opportunity nexus		
Sep 16	2. <i>Clinic: ideation and creativity</i>		
Sep 23	3. Opportunities from technology		
Sep 30	4. Opportunities for society		
Oct 7	5. Presentations: opportunity set	Gallo & Conti (Startup Launchpad), Lattuada (Technology Transfer Office), Delisle (SPOT) NEW!	1. Identification (weeks 1-5)
Oct 14	6. Validation I: scientific approach		
Oct 21	--- no session ---		
Oct 28	7. Validation II: industry analysis		
Nov 4	8. <i>The uncomfortable truth</i>	Vincent Bieri (Nextthink co-founder )	2. Validation (weeks 6-9)
Nov 11	9. Feedback session: validation review	Student founders (tbc)	
Nov 18	10. Business models I: creating value		
Nov 25	11. <i>Field trip: EPFL Investor Day</i>		
Dec 2	12. Business models II: capturing value		
Dec 9	13. <i>Clinic: pitching and deck design</i>		
Dec 16	14. Presentation: venture concepts	Giorgia Scetta (MassChallenge), Cecilia Molina (TalentKick), Mike Lamotte (InnovationPark), others tbc	3. Business design (weeks 10-14)

Any
volunteers?

NEW!

... to be continued (MGT-436)

Your journey

Date	Topics	
Sep 9	1. Founder-opportunity nexus	
Sep 16	2. <i>Clinic: ideation and creativity</i>	
Sep 23	3. Opportunities from technology	
Sep 30	4. Opportunities for society	
Oct 7	5. Presentations: opportunity set	Venture concepts posters (.pdf) by Oct 6 at noon (<i>individual</i>)
Oct 14	6. Validation I: scientific approach	
Oct 21	--- no session ---	
Oct 28	7. Validation II: industry analysis	
Nov 4	8. <i>The uncomfortable truth</i>	
Nov 11	9. Feedback session: validation review	Validation review deck (.pptx) by Nov 10 at noon (<i>team</i>)
Nov 18	10. Business models I: creating value	
Nov 25	11. <i>Field trip: EPFL Investor Day</i>	
Dec 2	12. Business models II: capturing value	
Dec 9	13. <i>Clinic: pitching and deck design</i>	
Dec 16	14. Presentation: venture concepts	Venture concept deck (.pptx) by Dec 15 at noon (<i>team</i>)

Deliverables

A. New Venture Project

Three submissions & three presentations
individual & team

Your journey

Date	Topics	
Sep 9	1. Founder-opportunity nexus	
Sep 16	2. <i>Clinic: ideation and creativity</i>	
Sep 23	3. Opportunities from technology	
Sep 30	4. Opportunities for society	
Oct 7	5. Presentations: opportunity set	Venture concepts posters (.pdf) by Oct 6 at noon (<i>individual</i>)
Oct 14	6. Validation I: scientific approach	<i>Team rosters due on Oct 13 at noon</i>
Oct 21	--- no session ---	
Oct 28	7. Validation II: industry analysis	
Nov 4	8. <i>The uncomfortable truth</i>	
Nov 11	9. Feedback session: validation review	Validation review deck (.pptx) by Nov 10 at noon (<i>team</i>)
Nov 18	10. Business models I: creating value	<i>Team rosters due on Nov 17 at noon</i>
Nov 25	11. <i>Field trip: EPFL Investor Day</i>	
Dec 2	12. Business models II: capturing value	
Dec 9	13. <i>Clinic: pitching and deck design</i>	
Dec 16	14. Presentation: venture concepts	Venture concept deck (.pptx) by Dec 15 at noon (<i>team</i>)

Deliverables

A. New Venture Project

Three submissions & three presentations
individual & team

Your journey

Date	Topics	
Sep 9	1. Founder-opportunity nexus	
Sep 16	2. <i>Clinic: ideation and creativity</i>	
Sep 23	3. Opportunities from technology	
Sep 30	4. Opportunities for society	
Oct 7	5. Presentations: opportunity set	Venture concepts posters (.pdf) by Oct 6 at noon (<i>individual</i>)
Oct 14	6. Validation I: scientific approach	+ Feedback forms (<i>individual</i>)
Oct 21	--- no session ---	
Oct 28	7. Validation II: industry analysis	
Nov 4	8. <i>The uncomfortable truth</i>	
Nov 11	9. Feedback session: validation review	Validation review deck (.pptx) by Nov 10 at noon (<i>team</i>)
Nov 18	10. Business models I: creating value	+ Feedback forms (<i>individual</i>)
Nov 25	11. <i>Field trip: EPFL Investor Day</i>	
Dec 2	12. Business models II: capturing value	
Dec 9	13. <i>Clinic: pitching and deck design</i>	
Dec 16	14. Presentation: venture concepts	Venture concept deck (.pptx) by Dec 15 at noon (<i>team</i>)
		+ Feedback forms (<i>individual</i>)

Deliverables

A. New Venture Project

Three submissions & three presentations
individual & team

B. Peer-to-Peer Feedback

Three feedback forms
individual

Your journey

Date	Topics	
Sep 9	1. Founder-opportunity nexus	
Sep 16	2. Clinic: ideation and creativity	
Sep 23	3. Opportunities from technology	
Sep 30	4. Opportunities for society	
Oct 7	5. Presentations: opportunity set	Venture concepts posters (.pdf) by Oct 6 at noon (<i>individual</i>)
Oct 14	6. Validation I: scientific approach	+ Feedback forms (<i>individual</i>)
Oct 21	--- no session ---	
Oct 28	7. Validation II: industry analysis	
Nov 4	8. The uncomfortable truth	
Nov 11	9. Feedback session: validation review	Validation review deck (.pptx) by Nov 10 at noon (<i>team</i>)
Nov 18	10. Business models I: creating value	+ Feedback forms (<i>individual</i>)
Nov 25	11. Field trip: EPFL Investor Day	
Dec 2	12. Business models II: capturing value	
Dec 9	13. Clinic: pitching and deck design	
Dec 16	14. Presentation: venture concepts	Venture concept deck (.pptx) by Dec 15 at noon (<i>team</i>)
		+ Feedback forms (<i>individual</i>)
		Reflection note (.pdf) by Dec 22 at noon (<i>individual</i>)

Deliverables

A. New Venture Project

Three submissions & three presentations
individual & team

B. Peer-to-Peer Feedback

Three feedback forms
individual

C. Reflection note

Short report (max. 1000 words)
individual

Your journey

Date	Topics	
Sep 9	1. Founder-opportunity nexus	
Sep 16	2. Clinic: ideation and creativity	
Sep 23	3. Opportunities from technology	
Sep 30	4. Opportunities for society	
Oct 7	5. Presentations: opportunity set	Venture concepts posters (.pdf) by Oct 6 at noon (<i>individual</i>)
Oct 14	6. Validation I: scientific approach	+ Feedback forms (<i>individual</i>)
Oct 21	--- no session ---	
Oct 28	7. Validation II: industry analysis	
Nov 4	8. The uncomfortable truth	
Nov 11	9. Feedback session: validation review	Validation review deck (.pptx) by Nov 10 at noon (<i>team</i>)
Nov 18	10. Business models I: creating value	+ Feedback forms (<i>individual</i>)
Nov 25	11. Field trip: EPFL Investor Day	
Dec 2	12. Business models II: capturing value	
Dec 9	13. Clinic: pitching and deck design	
Dec 16	14. Presentation: venture concepts	Venture concept deck (.pptx) by Dec 15 at noon (<i>team</i>)
		+ Feedback forms (<i>individual</i>)
		Reflection note (.pdf) by Dec 22 at noon (<i>individual</i>)

Deliverables

A. New Venture Project (60%)

Three submissions & three presentations
individual & team

B. Peer-to-Peer Feedback (20%)

Three feedback forms
individual

C. Reflection note (20%)

Short report (max. 1000 words)
individual

A real journey



Umami Threads
Timeless apparel basics at a fair price

Business plan presentation
SSE - 1203 Business Plan, Fall 2014

August Bard Brिंगéus
Jakob Nilsson Dworsky
Davide Bavato
Michel Greter

Fall 2014



Umami Threads

Timeless apparel basics at a fair price

Business plan presentation
SSE - 1203 Business Plan, Fall 2014

August Bard Bringéus
Jakob Nilsson Dworsky
Davide Bavato
Michel Greter



Fall 2014



The ASKET T-shirt

 [Project We Love](#)  [Stockholm, Sweden](#)  [Apparel](#)

€45,795

pledged of €10,000 goal

674

backers



Spring 2015



Spring 2018



Today

PART II

The most entrepreneurial nation

E'ship trivia

Which nation has the highest concentration of entrepreneurs?

E'ship trivia

Which nation has the highest concentration of entrepreneurs?

Percentage of self-employed / independent workers

World Bank, 2023

Chad	91.6
Niger	89.5
Sierra Leone	88.8
Madagascar	88.3
Nigeria	86.2
...	
Switzerland	16.9
European Union	14.2
US	6.1



E'ship trivia

Which nation has the highest concentration of entrepreneurs?

Percentage of students wishing to establish a company
GUESSSS

	End of studies (%)	5y after (%)
Uganda	51.5	70.7
Argentina	55.0	70.0
Tanzania	47.4	66.7
AVERAGE	15.4	28.3 ↓
...		
Switzerland	5.9	20.2
United States	14.3	34.0

Entrepreneurship

A (very) broad definition

The activity of **setting up a business** or businesses,
taking on financial risks in the hope of profit.

from Oxford Dictionary of English

Different shades of entrepreneurship

- ❑ **Forced entrepreneur:** low capital requirements, limited R&D, small scale, necessity or sustenance



Different shades of entrepreneurship

- ❑ **Forced entrepreneur:** low capital requirements, limited R&D, small scale, necessity or sustenance

The Journal of FINANCE

The Journal of THE AMERICAN FINANCE ASSOCIATION

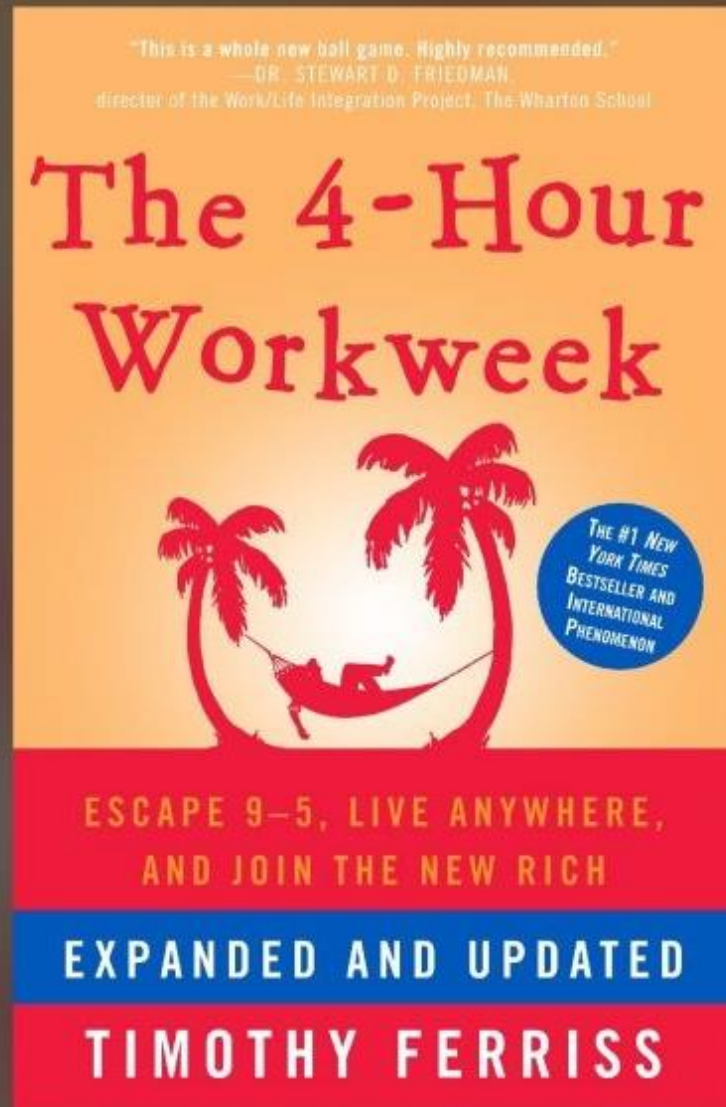
THE JOURNAL OF FINANCE • VOL. LXXVII, NO. 1 • FEBRUARY 2022

Forced Entrepreneurs

ISAAC HACAMO and KRISTOPH KLEINER

ABSTRACT

Conventional wisdom suggests that labor market distress drives workers into temporary self-employment, lowering entrepreneurial quality. Analyzing employment histories for 640,000 U.S. workers, we document that graduating college during a period of high unemployment does increase entry to entrepreneurship. However, compared to voluntary entrepreneurs, firms founded by *forced* entrepreneurs are more likely to survive, innovate, and receive venture backing. Explaining these results, we confirm that labor shocks disproportionately impact high earners, with these workers starting more successful firms. Overall, we document untapped entrepreneurial potential across the top of the income distribution and the role of recessions in reversing this missing entrepreneurship.



Different shades of entrepreneurship

- ❑ **Forced entrepreneur:** low capital requirements, limited R&D, small scale, necessity or sustenance
- ❑ **Life-style business:** low capital requirements, small scale, self-achievement, autonomy

Different shades of entrepreneurship

- ❑ **Forced entrepreneur:** low capital requirements, limited R&D, small scale, necessity or sustenance
- ❑ **Life-style business:** low capital requirements, small scale, self-achievement, autonomy
- ❑ **Social venture:** low-to-moderate capital requirements, social or environmental purpose in addition to financial objectives

ROCKET
GRAN



RUN FORREST, RUN!
Babypullover Smiley



TIME FOR ACTION

Find out how we
can remain competitive
as a nation.

■ **Deep tech startup:** high capital requirements, lengthy R&D, aimed at developing and/or commercializing breakthrough technologies

Different shades of entrepreneurship



- ❑ **Forced entrepreneur:** low capital requirements, limited R&D, small scale, necessity or sustenance
- ❑ **Life-style business:** low capital requirements, small scale, self-achievement, autonomy
- ❑ **Social venture:** low-to-moderate capital requirements, social or environmental purpose in addition to financial objectives
- ❑ **Deep tech startup:** high capital requirements, lengthy R&D, aimed at developing and/or commercializing breakthrough technologies
- ❑ **Intrapreneurship:** low-to-high initial capital investment, company-dependent

Different shades of entrepreneurship



- ❑ **Forced entrepreneur:** low capital requirements, limited R&D, small scale, necessity or sustenance
- ❑ **Life-style business:** low capital requirements, small scale, self-achievement, autonomy
- ❑ **Social venture:** low-to-moderate capital requirements, social or environmental purpose in addition to financial objectives
- ❑ **Deep tech startup:** high capital requirements, lengthy R&D, aimed at developing and/or commercializing breakthrough technologies
- ❑ **Intrapreneurship:** low-to-high initial capital investment, parent company-dependent, low personal risk

Different shades of entrepreneurship

Types of entrepreneurship can vary by:

- ❑ Personal motive or organizational purpose
- ❑ Capital requirements
- ❑ Innovativeness
- ❑ Scalability
- ❑ Risk profile
- ❑ Time-to-market

PART III

The most valuable startups

E'ship trivia

Which are the most valuable startups in the world?



E'ship trivia

Which are the most valuable startups in the world?

Most valuable unicorns (privately held, valued >\$1 bn)

CB insights, Sep 2026

Company	Valuation (US\$bn)	City	Country
Anthropic	965	San Francisco	US
OpenAI	840	San Francisco	US
ByteDance	480	Beijing	China
Stripe	159	San Francisco	US
Databricks	100	San Francisco	US
VAST Data	80	New York	US
Revolut	75	London	UK
SHEIN	66	Singapore	Singapore
DeepSeek	50	Hangzhou	China

Entrepreneurial drivers

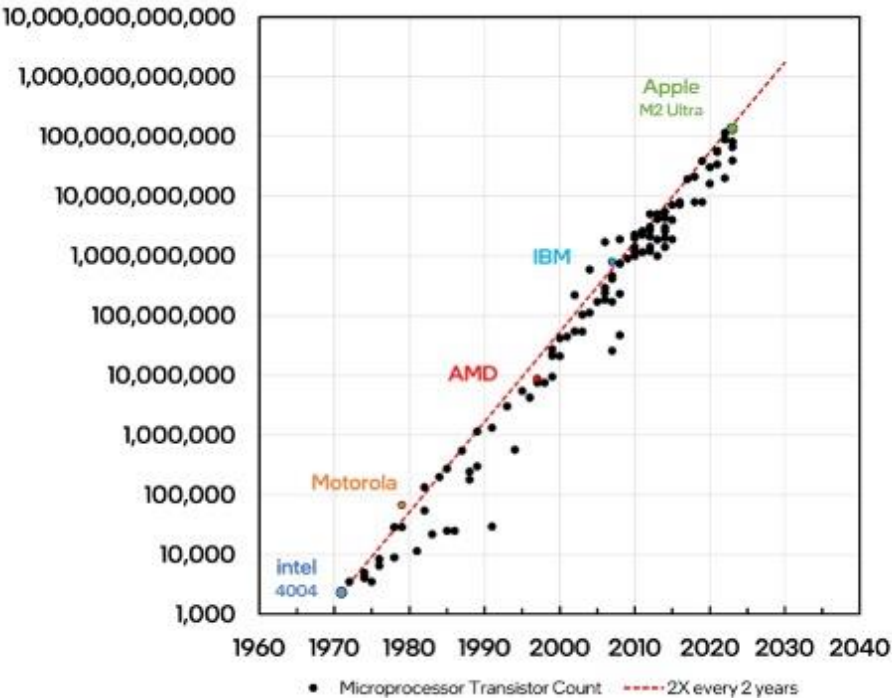
Technology

Paradigm	Period	Firms
Mainframe computing & telecommunications	1960s–1970s	IBM, AT&T, DEC
Microprocessor & personal computing	1970s–1990s	Intel, Microsoft, Apple, IBM
Internet & networking	1990s–2000s	Cisco, Microsoft, Google, Amazon
Mobile, cloud & digital economy	2000s–2010s	FAANG
AI & advanced semiconductors	2020s– ?	MANGOS (?)



Entrepreneurial drivers

Technology

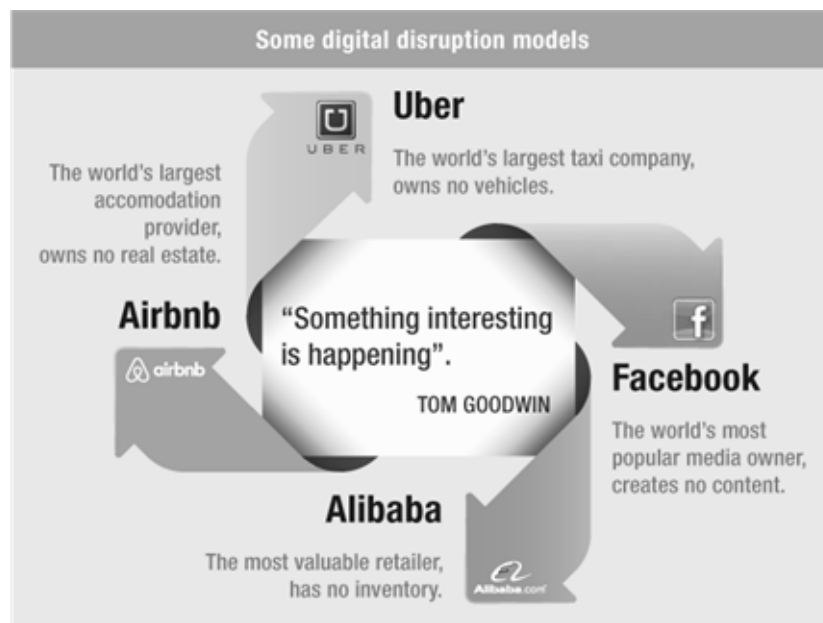


Paradigm	Period	Firms
Mainframe computing & telecommunications	1960s–1970s	IBM, AT&T, DEC
Microprocessor & personal computing	1970s–1990s	Intel, Microsoft, Apple, IBM
Internet & networking	1990s–2000s	Cisco, Microsoft, Google, Amazon
Mobile, cloud & digital economy	2000s–2010s	FAANG
AI & advanced semiconductors	2020s– ?	MANGOS (?)

Ranade, P., & Keller, J. (2023). *The relentless pursuit of Moore's Law: Part 1—Abstraction layers*. Bits and Bytes, July 16.



MASTERWORKS



Source: Tom Goodwin, Havas

Entrepreneurial drivers

Business models

Business paradigm	Logic	Firms
Mass production	Standardization and scale	Ford
Franchising	Product is the model	McDonald's
E-commerce	Cut the middleman	Amazon, Dell, Alibaba
Platforms & marketplaces	Create the middleman	Facebook, Airbnb, Uber
Everything-as-a-Service vs productized services	Products as services, services as products	AWS, TurboTax
Tokenization & decentralized models	Decentralize ownership / coordination	Masterworks, OpenSea

Entrepreneurship

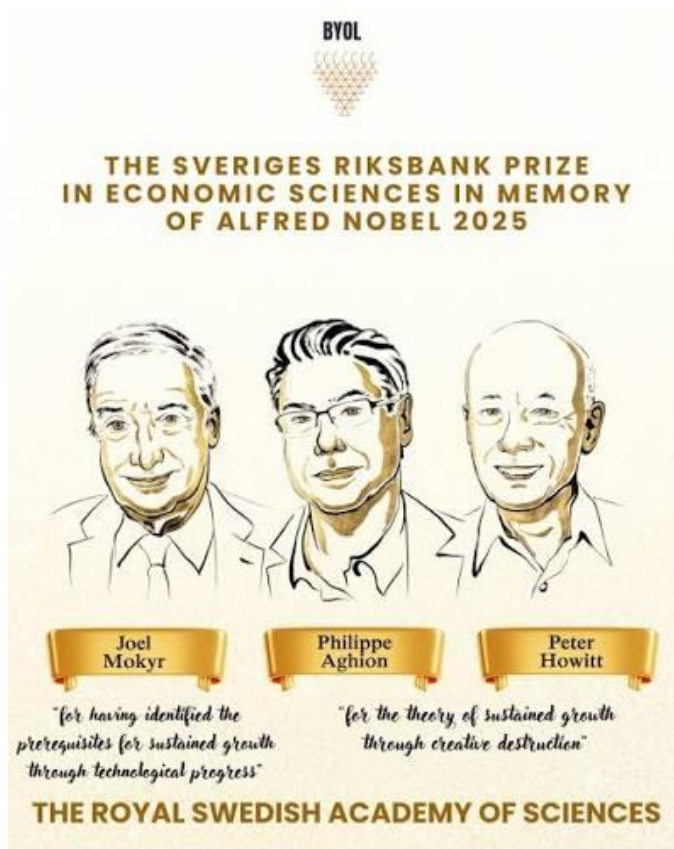
A (quite) old perspective

Entrepreneurship is a fundamental impulse of creative destruction, displacing existing structures through new technological-economic configurations.

after Joseph Schumpeter, *Capitalism, Socialism and Democracy* (1942)

Entrepreneurship

but still relevant



Entrepreneurship is a fundamental impulse of **creative destruction**, displacing existing structures through new **technological-economic configurations**.

after Joseph Schumpeter, *Capitalism, Socialism and Democracy* (1942)

Entrepreneurial drivers



Entrepreneurial drivers



VC Ecosystem
(Pitchbook, 2019-2025)

San Francisco
New York
Los Angeles*
Beijing
Shanghai*
Boston
London

UNPACK THE DATA

Select a measure to view startup activity across the world map. Measures can either be a level of activity (such as the amount of capital invested) or a change over time (such as percentage change in capital invested). The round of investment and time period can also be selected. But first, you must select one of either venture deals or capital invested. Deals has been pre-selected for you.

VENTURE ACTIVITY

CAPITAL INVESTED (\$)

VENTURE MEASURE

CAPITAL INVESTED (\$ M) PER 1M RESIDENTS

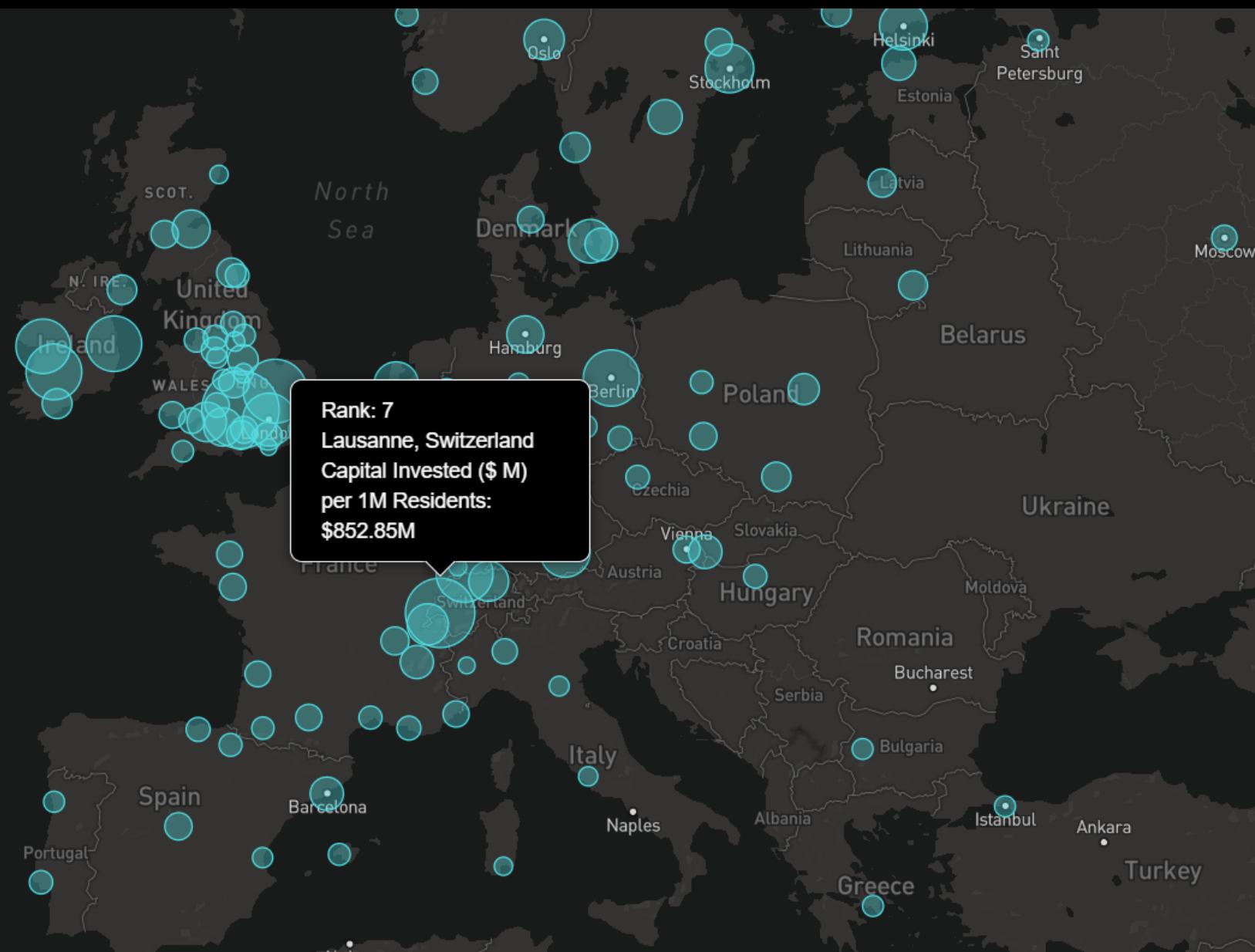
VENTURE ROUND

EARLY-STAGE VC

VIEW HISTORICAL DATA

2015-2017

Note: Population data only available for the 2015-17 period.



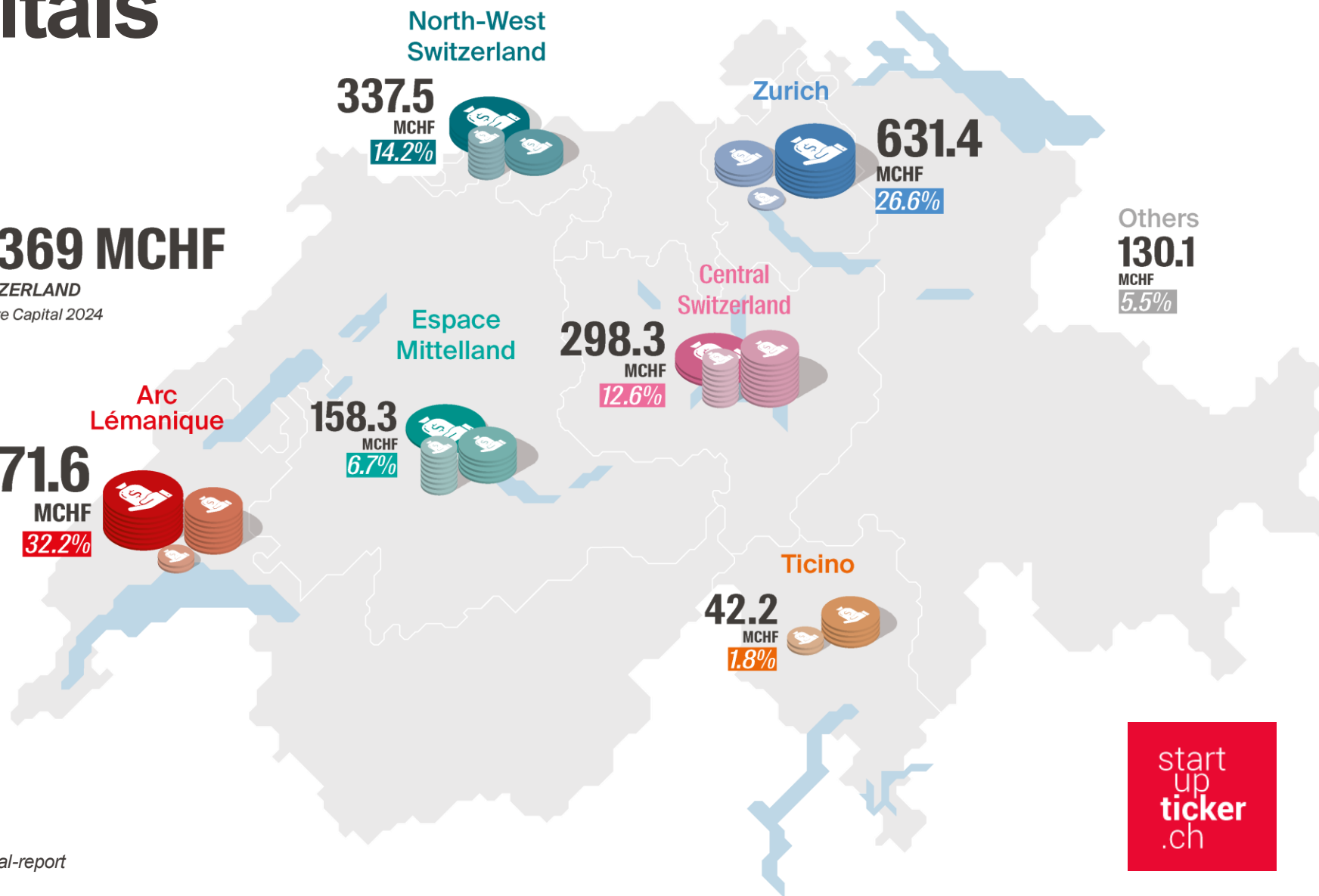
Venture Capitals 2024

**2'369 MCHF**SWITZERLAND
Venture Capital 2024

EPFL startups
have raised

701 MCHF

in 2025



Unlocking ecosystems synergies

EPFL

ACCELERATION PROGRAMS

EPFL
Innovation ParkTECH4
GROWTH
INDUSTRY4.0Future of
Health GrantTECH4
GROWTH
HEALTHTECH4
EVATrust4SMEs
Cybersécurité et confiance
numérique pour PME

TECH4REGENERATION

TECH4
TRUST

Innovation Parks, clusters

Microcity | Campus Biotech
Energypolis | Smart Living Lab
Biopôle | Impact Hub | Bioalps
Alliance | Think Sport | Unlimitrust

REGIONAL LEVEL

Coaching/financing

FIT | Innovaud
Venturekick | PERL
Business Angels Switzerland
Mass Challenge

Academics, Research

UNIL | CHUV | IMD
EHL | CSEM

NATIONAL LEVEL

Startup competitions

Venture Competitions
ESA BIC
EY entrepreneur of the
year
TOP100 startup
De Vigier

Startups exposure and events

Startup DAYS
START Summit

PART IV

Next steps

Join the discord server



DISCORD

go.epfl.ch/TV26_discord

